Ayoub Tarek

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EDUCATION

IMT Nord Europe, École Mines-Télécom – Université de Lille — Douai / Lille, France
Engineering Degree – Artificial Intelligence & Data Science track — *Sep 2024 – Expected 2027*

- ***Relevant Coursework:*** Machine Learning, Deep Learning, Time Series, Images Processing Fundamentals, Optimisation, Data Mining, Big Data Architectures, Research Discovery Project.

Lycée Lavoisier, Lycée Raspail — Paris, France
Scientific Preparatory Classes – PCSI/PSI — *Sep 2022 – Jul 2024*

- Intensive two-year program in Mathematics, Physics and Chemistry for competitive entry to engineering schools.

RESEARCH EXPERIENCE

GE HealthCare – Interventional Image Processing Research Team — Buc, France
AI Research Intern — *May 2026 – Aug 2026*

- Investigated probe pose estimation by benchmarking vision foundation models (RAD-DINO, FluoroSAM, DINOv2) and in-house CNNs against a production baseline via structured evaluation and full/LoRA fine-tuning.
- Adapted an adversarial unsupervised domain adaptation method (FARR) to pose regression with fine-tuned DINOv2, reducing the gap between two probe configurations without target labels; tuned with Optuna.

Spiking Neural Networks for Drone Detection | ***Group Research Project (IMT Nord Europe)*** — *Sep 2025 – Apr 2026*

- Contributed, as part of a student research team, to a comparative study of spiking neural network (SNN) architectures against a standard ANN baseline for drone detection on FRED (Florence RGB-Event Drone Dataset), an event-camera benchmark.
- Helped implement and train SpikeYOLO (BICLab) and a custom SNN-YOLOv1 (snnTorch), alongside a fine-tuned YOLOv8n baseline, exploring the trade-offs between spiking and conventional architectures for event-based vision. (GitHub)

COMPETITIONS

OncoVIA – 1st Prize | ***Unboxed AI & Health Hackathon, Centrale Lyon*** — *Feb 2026*

- Built an AI-assisted oncology diagnostic tool combining automated CT-scan analysis with a Mistral-AI agent correlating textual medical history and 3D imaging to generate structured medical reports.
- Designed the end-to-end pipeline, from image preprocessing to report generation, as a fast and explainable diagnostic-support workflow.
- Continuing to develop the project beyond the hackathon. (GitHub)

Surgical JEPA | ***EB-JEPA Hackathon (HackTheWorld(s))*** — *Jun 2026*

- Adapted EB-JEPA (Energy-Based Joint-Embedding Predictive Architecture) into an action-conditioned world model for surgical robotics, using wrist-camera video from the NVIDIA Open-H-Embodiment surgical suturing dataset.
- Designed a causal, action-conditioned Transformer predictor (AdaLN-Zero conditioning) and a DINOv3-ConvNeXt encoder variant, and trained an RGB decoder on frozen JEPA latents for qualitative evaluation of rollouts.
- Evaluated design choices through LPIPS-based perceptual comparison of autoregressive rollouts. (GitHub)

PROJECTS & OTHER EXPERIENCE

Excelya — Boulogne-Billancourt, France
AI Intern – Applied Healthcare AI — *May 2025 – Aug 2025*

- Built and deployed a local Retrieval-Augmented Generation (RAG) pipeline for document processing, covering LLM integration, embeddings, prompt design and vector stores (FAISS, ChromaDB).
- Designed the retrieval and indexing pipeline to run fully on-premise, without relying on external API calls.

Music Information Retrieval & Machine Learning | ***Course Project (IMT Nord Europe)*** — *Feb 2026*

- Built genre classification and popularity regression models from Spotify audio features, comparing KNN, Logistic Regression, SVM, Random Forest and XGBoost, and designed a KNN-based track recommender with conformal prediction to quantify uncertainty.
- Implemented a Shazam-style audio fingerprinting pipeline: STFT spectrograms, spectral peak extraction, combinatorial hashing and time-aligned hash matching to identify tracks from short excerpts.

SKILLS & INTERESTS

Languages: French (Native), English (C1, TOEFL 110), Arabic (Fluent)

Programming: Python, SQL, JavaScript

ML/DL & Research Tools: PyTorch, Hugging Face (Transformers, PEFT/LoRA), Optuna, TensorBoard, snnTorch, scikit-learn, XGBoost, librosa, FAISS, ChromaDB

Interests: Healthcare, Fashion, Golf, Basketball